Adrenaline (Epinephrine) – for HYPOTENSION

MORE THAN ONE STRENGTH AVAILABLE – ENSURE THE CORRECT STRENGTH IS USED FOR THE APPROPRIATE INDICATION

Neonatal

FORM Adrenaline 1:1000 (1mg/1ml)
(IM preparations can be given intravenously if diluted as below)

INDICATION Refractory hypotension

Refractory hypotension

DOSE RANGE

AGE	DOSE RANGE	FREQUENCY	ROUTE
Birth - 6 months	0.01 – 0.1 microgram/kg/minute Doses of up to 0.3micrograms/kg/minute may be required in exceptional circumstances under consultant advice.	Continuous infusion	IV

NB: For this indication Adrenaline 1:1000 (1mg/1ml) strength is required. Special care is required when using and storing this strength of adrenaline.

PRESCRIPTION OF CONTINUOUS INFUSION

LOW CONCENTRATION - STARTING CONCENTRATION

FOR DOSES ≤ 0.1microgram/kg/minute
0.3mg/kg in 50ml infusion fluid
This gives:-

- 0.02 microgram/kg/minute at 0.2ml/hour
- 0.05 microgram/kg/minute at 0.5ml/hour
- 0.1 microgram/kg/minute at 1ml/hour

HIGH CONCENTRATION

FOR DOSES > 0.1microgram/kg/minute
3mg/kg in 50ml infusion fluid
This gives:-

- 0.1 microgram/kg/minute at 0.1 ml/hour
- 0.3 microgram/kg/minute at 0.3 ml/hour

RECONSTITUTION Already in solution. **1:1000 strength must be diluted prior to use.**

DILUTION

LOW CONCENTRATION
Using Adrenaline 1 in 1000 (1mg/1ml) injection;
0.3 x weight (kg) is the number of ml of adrenaline 1mg/1ml to be diluted up to 50ml total with infusion fluid (equivalent to 0.3mg/kg in 50ml)

HIGH CONCENTRATION
Using Adrenaline 1 in 1000 (1mg/1ml) injection;
3 x weight (kg) is the number of ml of adrenaline 1mg/1ml to be diluted up to 50ml total with infusion fluid (equivalent to 3mg/kg in 50ml)

METHOD OF ADMINISTRATION

For continuous Infusion
By continuous intravenous infusion, flow rate adjusted according to the baby's response (see prescription section for details).

RHC – NB If the infusion pump alarms for breach of a hard dose limit, the pump can be reprogrammed using the 'dosing only option' if switching to high concentration is not clinically appropriate for dose titration.

COMPATIBILITY

Solution compatibility	Sodium chloride 0.9%, Glucose 5%
Solution incompatibility	No information
IV Line compatibility	Amikacin, Amiodarone (in glucose only), Caffeine citrate, Calcium Gluconate , Dobutamine, Dopamine, Fentanyl, Furosemide (in sodium chloride only), Hydrocortisone, Insulin, Labetalol, Levofloxacin, Midazolam, Milrinone, Morphine, Noradrenaline, Potassium Chloride, Vecuronium
IV Line incompatibility	Omeprazole, Sodium bicarbonate

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
See Summary of Product Characteristics and most recent edition of BNF for Children

FURTHER INFORMATION

Monitoring requirements:
Heart rate and rhythm, arterial blood pressure, peripheral perfusion and blood glucose.
Monitor urine output and renal function.

Ideally should be administered via central line. Risk with extravasation resulting in tissue damage/necrosis if given peripherally as adrenaline is a potent vasoconstrictor and has a low pH

STORAGE

Do not use discoloured solutions or solution containing a precipitate.

PH

2.5 – 4

LICENSED STATUS

Off label use in neonates. IM adrenaline 1:1000 is licensed to be administered IV when diluted to concentrations of 0.1mg/ml.

NEONATAL West of Scotland Drug Monographs
Parenteral Route

APPLICABLE POLICIES West of Scotland Neonatal Guidelines
Consult local policy if applicable

Document Number:	007	Supersedes:	006
Prepared by/Checked by:	Jacqueline Rooney / Andrew Stockton/Roisin McCabe	Approved by:	WOS Neo Pharm Group
Date prepared	August 2026	Review Date	August 2029

Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department